

COURSE TITLE : MANUFACTURING AND PRESS TOOL TECHNOLOGY
COURSE CODE : 6301
COURSE CATEGORY : A
PERIODS / WEEK : 5
PERIODS / SEMESTER : 75
CREDIT : 5

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Foundry and Casting	18
2	Welding soldering and brazing.	18
3	Metal cutting and mechanical working of metals	18
4	Press Tools	21
	TOTAL	75

COURSE OUTCOME:

On completion of the study the students will be able to:

1. Comprehend the foundry and casting processes.
2. Comprehend the welding, soldering and brazing
3. Understand the various aspects of metal cutting
4. Understand the mechanical working of metals
5. understand the press tool parts and its operations

MODULE I

1.1.0 Comprehend the foundry and casting processes

- 1.1.1 Explain the necessity of making different types of pattern
- 1.1.2 List the various pattern material and state the advantages and disadvantages
- 1.1.3 List and explain various pattern allowances and scales
- 1.1.4 List and explain properties of moulding sand
- 1.1.5 List the types of moulding sand
- 1.1.6 List the ingredients of sand
- 1.1.7 Mention the uses of chaplets
- 1.1.8 Explain the different methods of moulding
- 1.1.9 Identify the different moulding operations
- 1.1.10 Explain the pouring techniques of molten metal
- 1.1.11 Mention the function of runner and risers
- 1.1.12 State the importance of permanent mould casting and mention its limitation
- 1.1.13 List the cleaning methods of casting and explain it
- 1.1.14 List the special types of casting
- 1.1.15 List the steps of die casting processes
- 1.1.16 State the processes of gravity die casting and pressure die casting
- 1.1.17 Explain the process of slush casting

MODULE II

2.1.0 Comprehend the welding, soldering and brazing.

- 2.1.1 Define welding and types of welding, advantages and limitations of welding.
- 2.2.0 Illustrate the principle of arc welding and various joints and its defects
- 2.2.1 Discriminate D C generators and A C transformer
- 2.2.2 Identify the electrodes for a particular weld and the specification of electrodes
- 2.2.3 Define gas welding, Identify oxy -acetylene welding and the types of flame
- 2.2.4 State the functions of pressure regulators, welding torch and nozzles, filler rod and flux
- 2.2.5 Explain submerged arc welding, tungsten inert gas (TIG) welding, Metal inert gas (MIG) welding.
- 2.2.6 Classify soldering and brazing
- 2.2.7 Compare welding, soldering and brazing

Module III

3.1.0 Understand the various aspects of metal cutting

- 3.1.1 Draw the geometry of orthogonal cutting.
- 3.1.2 Explain the action of cutting tool by orthogonal cutting and oblique cutting.
- 3.1.3 Explain the chip formation with diagram.
- 3.1.4 Name the various types of chips.
- 3.1.5 Distinguish the effect of cutting speed, feed and depth of cut on cutting force.
- 3.1.6 Explain about the tool signature/ nomenclature of the single point cutting tool.
- 3.1.7 List the various cutting tool materials.
- 3.1.8 Compare the properties of various cutting tool materials
- 3.1.9 List the factors affecting the life of cutting tools
- 3.1.10 Illustrate the term machinability of a material.
- 3.1.11 Mention the energy efficiency in metal cutting
- 3.1.12 List the factors affecting energy efficiency the metal cutting

3.2.0 Understand the mechanical working of metals - cold working and hot working

- 3.2.1 Explain the cold working and hot working and its advantages and disadvantages
- 3.2.2 List the cold working operations
- 3.2.3 Explain the cold working operations and hot working operations
- 3.2.4 State the effects of hot working on metals

Module IV

4.1.0 Understand the working of Presses

- 4.1.1 List the parts of press.
- 4.1.2 State the classification of presses.
- 4.1.3 Illustrate the strip feed arrangements on press.

4.2.0 Understand Sheet Metal Processes using different types of Press Tools

- 4.2.1 List out the Press Tool operations.
- 4.2.2 Draw different strip layout.
- 4.2.3 Estimate percentage utilization and economy factor of strip.

4.3.0 Comprehend Punches, Dies and stoppers

- 4.3.1 List the type of punches.
- 4.3.2 State the type of die block.
- 4.3.3 Identify different type of stoppers.

4.4.0 Analyse the parts of Press Tool

- 4.4.1 Describe the functions of strippers.
- 4.4.2 State the dimensional aspect of gauges.
- 4.4.3 List the use of pilots and side cutters.
- 4.4.4 Compare ejectors and knockouts.

4.5.0 Know the progressive tool and Die sets

- 4.5.1 Calculate the shank location.
- 4.5.2 Identify the type of die sets.
- 4.5.3 List out the function of progressive tool.

4.6.0 Understand the working of compound tool

- 4.6.1 State the elements of compound tool.
- 4.6.2 Compare progressive and compound tool.

CONTENT DETAILS

MODULE I

Foundry and Casting-Types of patterns- single piece pattern- split pattern- match plate pattern- gated pattern- loose piece pattern- sweep pattern etc Pattern materials- wood- metal- plastics- plaster Advantages & Disadvantages of Pattern materials Pattern allowances- shrinkage allowance- draft allowance- machining allowance- distortion or camber allowance- rapping allowance Properties of molding sand- porosity- plasticity- adhesiveness- cohesiveness- refractoriness moulding sand: - Types Green sand- dry sand- parting sand- loam sand- facing sand- core sand- Composition of moulding sands- – their ingredient and use - uses of chaplets - Moulding processes: - Bench moulding- pit moulding- floor moulding and sweep moulding. Pouring and feeding – proper pouring techniques – functions of risers – importance of chiller - Castings – Types – Ferrous- Non ferrous-special castings-die casting – gravity die casting – pressure casting – goose neck type – direct injection type and cold chamber machine - Slush castings – vacuum casting – cleaning of casting – cut-off-trimming-shot and sand blasting.

MODULE II

Welding soldering and brazing Welding – introduction- classification of welding- advantages and limitation of welding. Arc welding – Principle of arc welding- arc welding machines – D.C. generators- A.C. transformers. Advantages and limitations of each. Welding positions — flat- horizontal - vertical and overhead welding- welded joints - butt- lap –corner-- tee--edge - V- joints- U-joint- symbols as per BIS code. welding electrodes- selection- Electrode coatings – functions of Electrode coating. Gas welding – Oxy – Acetylene welding- Major advantages and limitations – type of flames –functions and operation of oxy – acetylene cylinders- pressure regulators- welding torch- nozzle. Gas welding procedure – selection of filler rod and flux. Brief explanation of submerged arc welding- tungsten inert gas (TIG) welding- metal inert gas (MIG) welding- Atomic hydrogen welding. Thermite welding – brief description only. Defects in welding – causes and remedies of the defects porosity- poor penetration- warping- under cut- distortion crack- poor appearances. Testing and inspection of

welding joints (Name only): Soldering and brazing – brief description only. Comparison between welding-soldering and brazing

MODULE III

Metal cutting Orthogonal cutting and oblique cutting- chip formation -type of chips- cutting speed - feed and depth of cut - tool signature/ nomenclature of the single point cutting tool- 1.2.8 State the properties of various cutting tool materials- tool life problems- machinability- nomenclature of taper shank Twist drill- plain milling cutters. cutting tool material-lathe -drilling- milling- shaper. Cutting fluids Lubricants- coolants - requirement-properties- method of application- selection.

Cold working– Operations – Drawing- Squeezing Rolling- Bending- Shearing & Extruding – Hot Working – Operations – Rolling- Piercing- Drawing- Spinning & Extruding - Effects of Hot Working - Types – Advantages & Disadvantages

MODULE IV

Press tool parts, capacity, bed, slide, bolster plate, pitman, clutch, stroke, shut height, die space, capacity, number of slides, actuation of slide. Press frame types, clutch, brake, stroke adjustments. Classification of press – source of power – mechanical, hydraulic& pneumatic. Strip feed arrangements on press.

Press Tool operations Introduction to piercing tool, blanking tool, progressive tool, compound tool, cut off tool, parting tool, notching tool, trimming tool, shaving tool, performing tool, lancing tool, dinking tool, Side cam tool, broaching tool, extrusion tool, planishing tool, embossing tool, coining tool, bending tool, forming tool, drawing tool, curling tool, bulging tool, swaging tool, combination tool, assembly tool. Strip layout – narrow run-wide run - single row – double row – gang dies – grain direction — layout for blanking tool, progressive tool. Economy factor, calculation of percentage utilization of strip. Punches – cutting – non-cutting – Hybrid punches – types of punches – plain – pedestal – offset Pedestal– headless – step head – bevelled head– perforators - material of punch. Buckling of punches, buckling of theorem – problems – type of loads on punch, determining the size of punch. Die blocks – types – solid – split, sectional– selection of die block, Material for die block, Stoppers – functions – position – stop categories – primary – secondary final, Construction – pin stop – headed pin stop – disappearing pin stop – finger stop – pusher stop trigger stop.

Strippers – function, types-fixed and traveling. Constructional details – spring stripper, clamping spring, , spring around stripper bolt. Stripper bolt suspension, Stripping force for blanking & piercing Gauges (Strip guide), Functions – dimensional aspect of gauge, Types – back, front, bulge clearance extended back gauge – strip support – pushers, nesting gauges, fool proofing – strip lifters Pilots, Purpose, indirect and direct pilot , size, length of pilot, Pilot opening in die and in die shoe, Types – functions of different types of pilots, Methods of piloting Side cutters - Function - advantages Ejectors and Shedders, Function - types Knockouts, Function, - direct & indirect knockouts Fasteners and dowels, Dowel – function – types, Fasteners – screws – function – types of screws like socket head, counter sunk, cheese head, set screw fasteners, Rivets, cotter pins – positioning of screw progressive tool and die sets. Shank,Types, location- arithmetical and graphic method. Die sets, Types – precision and commercial / standard & nonstandard. Components – top plate – bottom plate, guide Pillar. Define Shut height, functions of Progressive tool. Construction and related design. Presses

Compound tool, Function of elements – related design, Progressive and compound tool – advantages disadvantages, Shaving – process – purpose, Clearance – punch and die size

TEXT BOOK

1. Work shop technology Vol- I, II - S. KHajraChoudhary,S.K.Bose, A.K. Hajra Choudhary, Nirjharroy
2. TOOL ENGINEERING & DESIGN- NAGPAL G R

REFERENCE BOOKS

John	Metal Casting and Joining
Mukherjee	Metal Fabrication Technology
Lindberg	Processes and Materials of Manufacture, 4th ed.
Er.R.K. Jain	Production technology
P.C. Sharma	Production Technology
Donaldson	Tool design